

System Documentation

Student Performance Prediction

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1. Abstract

Smart Analysis Reporter is a web-based statistical analysis and reporting toolkit. It loads a dataset, runs a battery of descriptive and inferential analyses, and assembles an editable, exportable analytics report. This document describes the system behind the toolkit and demonstrates it on the Student Performance Prediction dataset (10,000 rows × 8 columns), whose goal is to understand and predict student exam performance.

2. System Design

The application follows a layered, modular design. A thin data layer caches the dataset; a set of pure-Python analysis modules implement the statistics; a Streamlit multipage UI exposes them interactively; and a report engine turns any analysis into embeddable charts, tables and narrative that export to PDF, Word and HTML. An auto-report builder chains the modules to produce a full report in one click.

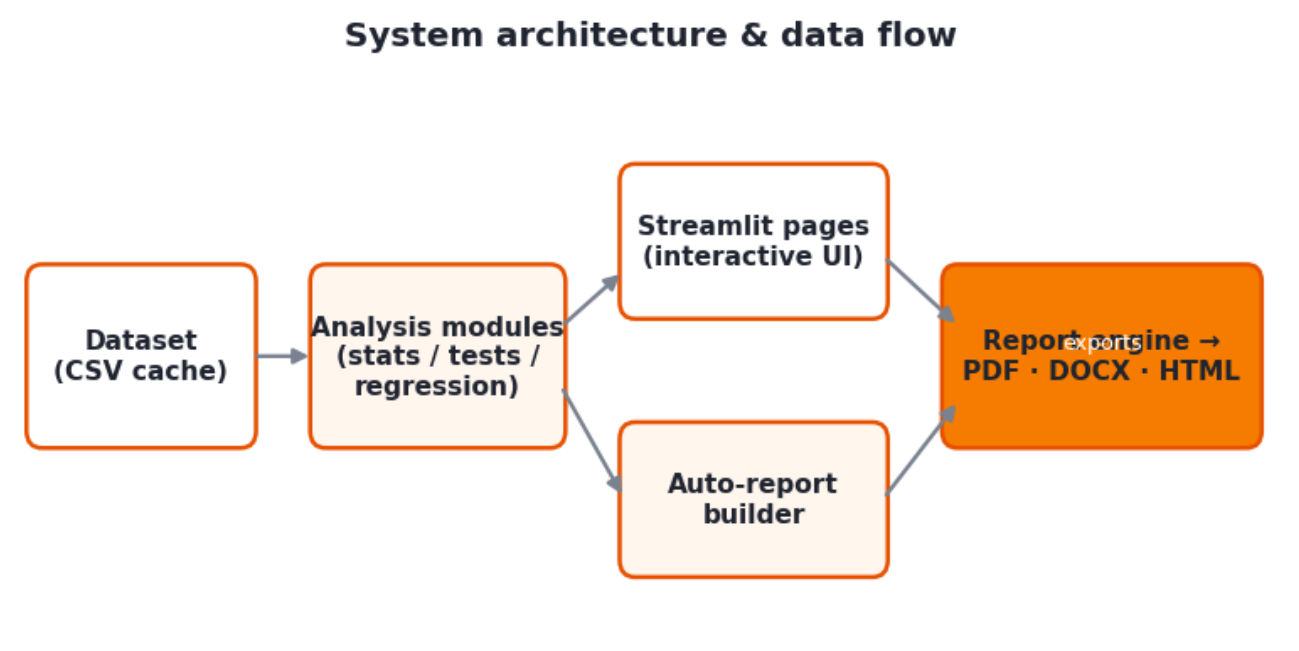


Figure 1 — System architecture and data flow.

3. Technology Stack

Layer	Technology	Purpose
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UI	Streamlit	Multipage interactive web app & widgets
Computation	pandas, NumPy	Data handling and vectorised maths
Statistics	SciPy, statsmodels	Tests, distributions, OLS regression
Interactive charts	Plotly	On-screen exploration
Report charts	Matplotlib	Themed images embedded in exports
Export	ReportLab, python-docx	PDF and Word generation
Data source	kagglehub	Fetching the demonstration dataset

4. Dataset Introduction

The Student Performance Prediction dataset contains 10,000 student records with 8 variables (7 metric, 1 categorical) and 0 missing values. The columns are:

Column	Type	Example
study_hours	Metric	7
attendance	Metric	56
sleep_hours	Metric	8
internet_usage	Metric	7
assignments_completed	Metric	10
previous_score	Metric	62
exam_score	Metric	100.0
placement_status	Categorical	Placed

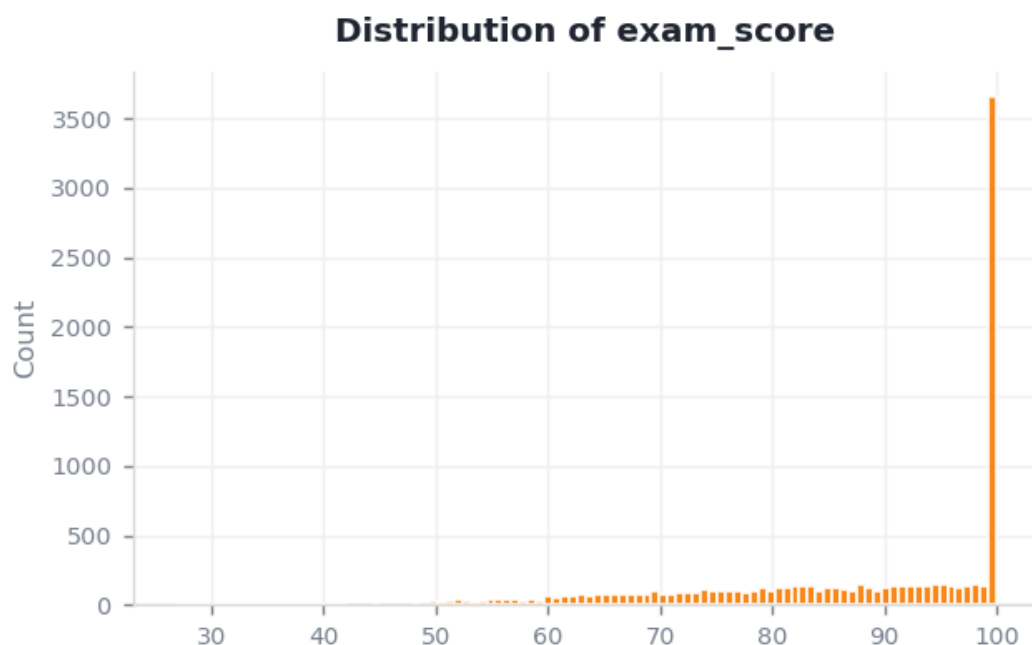


Figure 2 — Distribution of the target variable, exam_score.

5. Analysis

The toolkit applies descriptive statistics, frequency analysis, normality checks (Q-Q + Shapiro-Wilk), correlation, hypothesis tests (t, ANOVA, chi-square, Z) with live assumption checks, and linear regression. Two representative outputs are shown below.

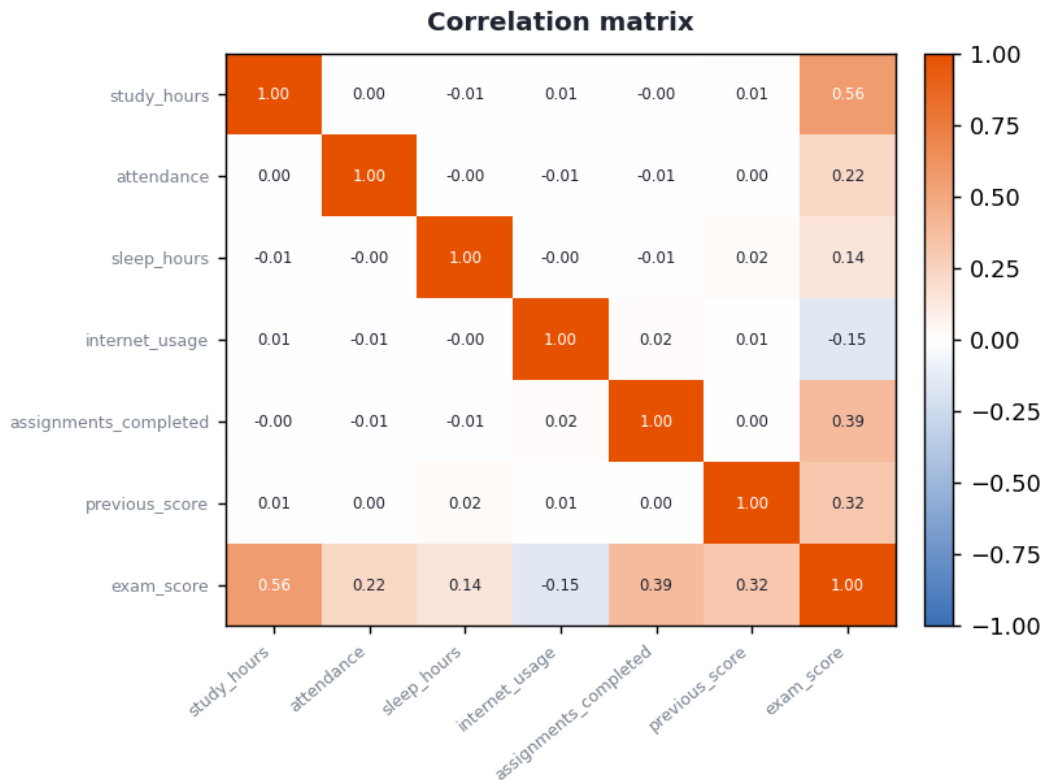


Figure 3 — Correlation matrix of the metric variables.

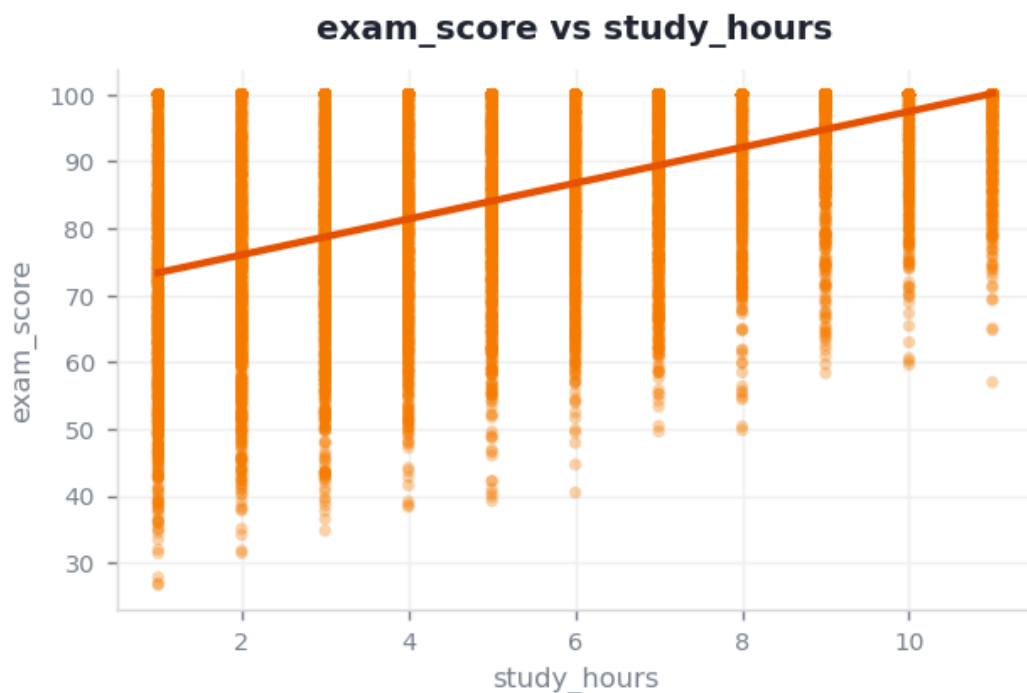


Figure 4 — Linear regression of exam_score on study_hours.

6. Conclusion of Analysis

- The strongest linear relationship is study_hours–exam_score ($r = 0.56$).
- A simple model of exam_score on study_hours explains 31.6% of the variance ($R^2 = 0.32$).
- There is a significant difference in exam_score between placement_status groups ($p < 0.001$).
- exam_score is non-normal (Shapiro-Wilk $W = 0.8376$).

7. Project Conclusion

Smart Analysis Reporter demonstrates an end-to-end analytics workflow: from raw data to an interactive toolkit to a polished, editable report — without writing code for each study. Its modular structure means new analyses or datasets can be added with minimal change, and its one-click reporting makes findings immediately shareable. The Student Performance Prediction case study shows the toolkit producing a complete, defensible statistical narrative automatically.